01 Full page - Era lanes

01-full-page-era-lanes.png

Default explorer with both pickers, layout controls, era toggles and the complete network.

Graph exploration

Semantic features (Jev): unavailable

All players · all teammate connections

Incomplete dated roster coverage, including 1946–1950 BAA evidence. Links require dated tenure overlap, an identified shared team game, or appearance counts that mathematically prove shared team games. Appearance proofs establish a relationship, but do not invent overlap dates or duration. Season-only membership, ambiguous counts and unresolved transaction brackets never create links. Degrees are exact within this evidenced graph; missing edges may shorten a historical chain or connect currently unreachable players.

Every player and every evidenced connection is loaded. Players are arranged by debut season; colors mark eras. Unknown debut is unclassified context. Select two players for an exact shortest teammate chain.

Player A

Any debut decade | Any franchise

Player B

Any debut decade | Any franchise

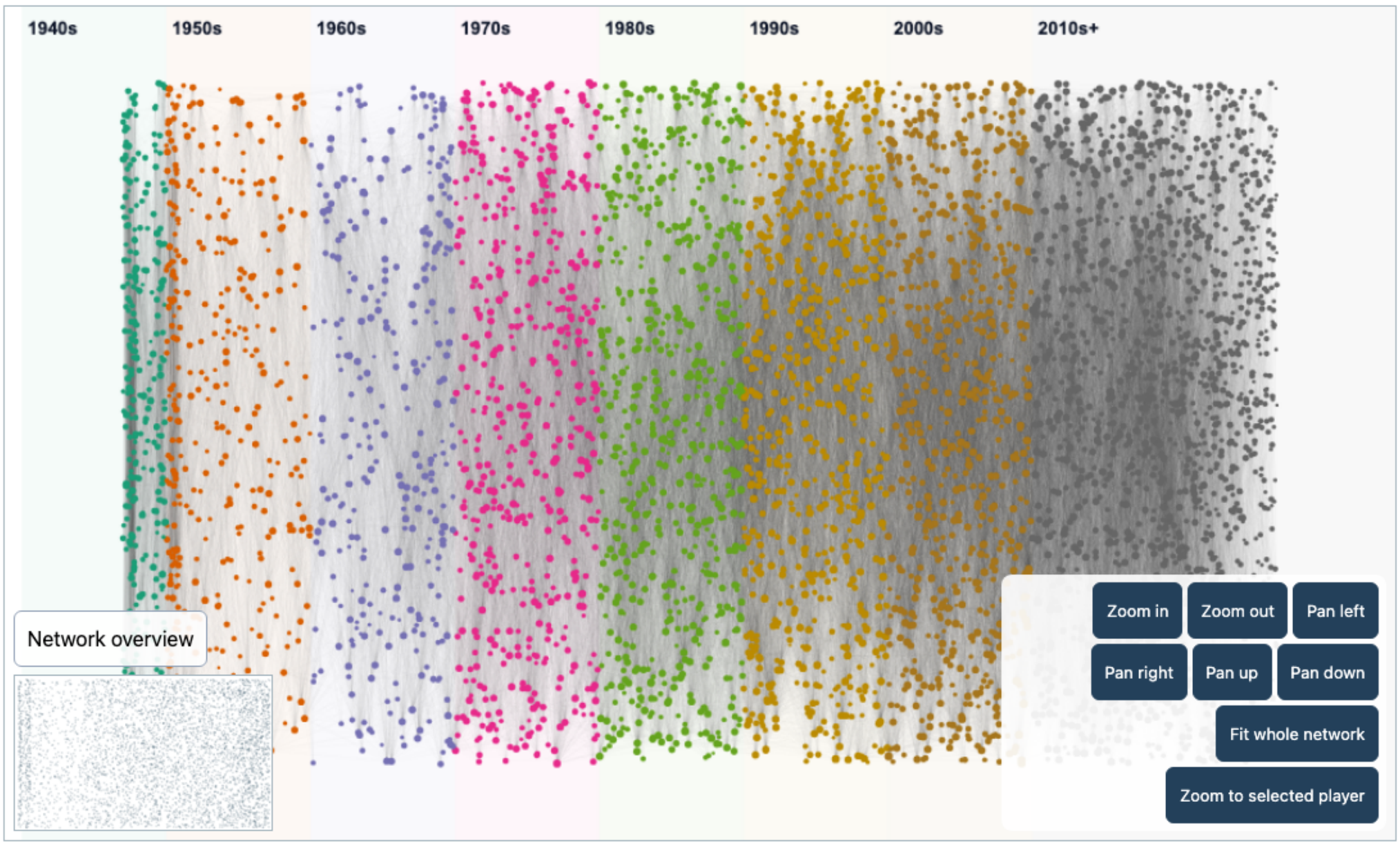
Connect players | Random pair | Clear chain

Era lanes layout | Concentric rings layout | Force-directed layout

1940s debut | 1950s debut | 1960s debut | 1970s debut | 1980s debut | 1990s debut | 2000s debut | 2010s+ debut

Show ego network only (1–2 hops) | Reset filters

All eras emphasized; franchise: all; 5106 matching of 5106 loaded players; 5106 visible. Whole network visible.



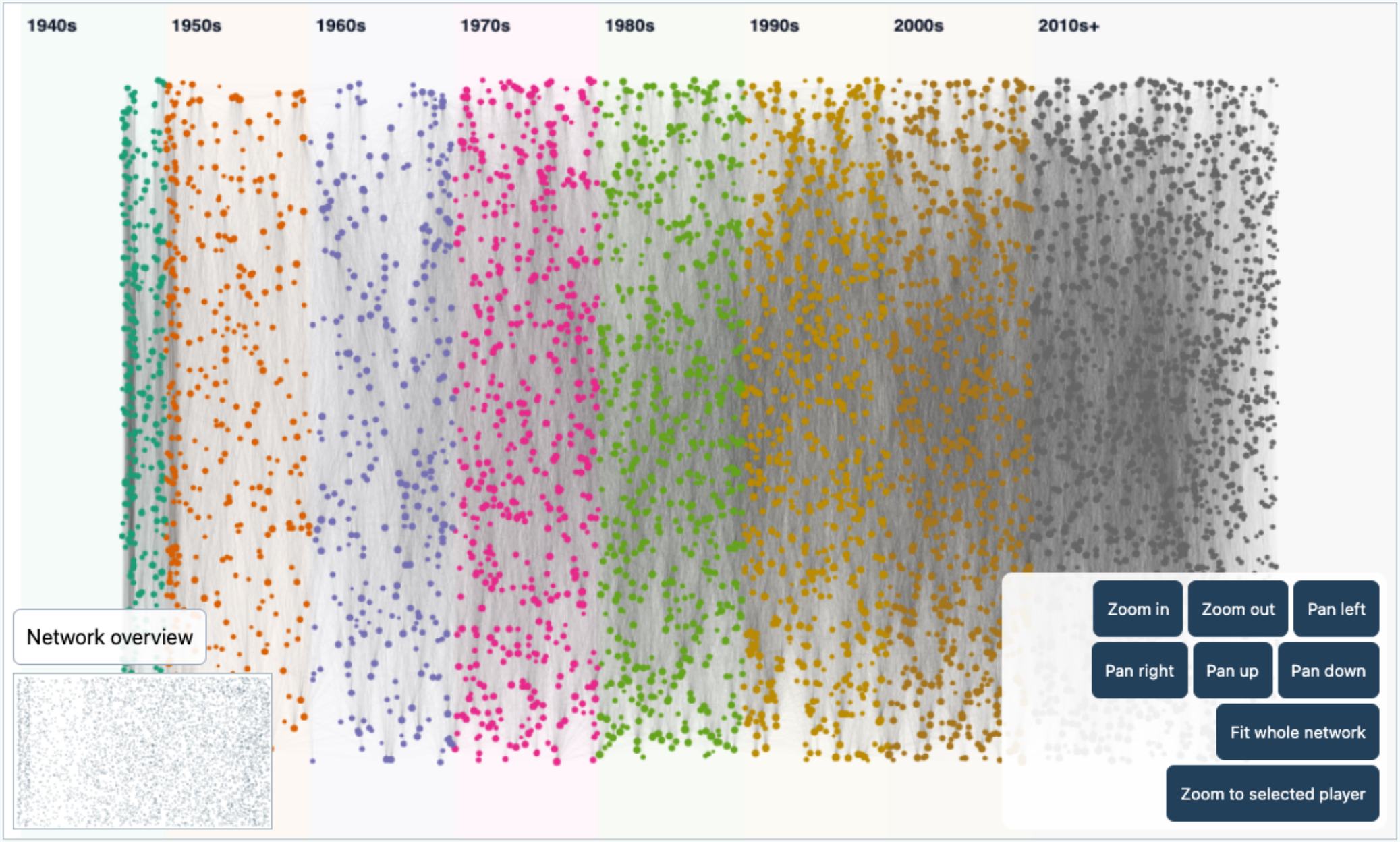
Drag to pan; scroll or pinch to zoom. Arrow keys pan; + and – zoom; f fits the network. Pick a player or click a relationship to inspect evidence.

Canvas ready — 5106 players, 101395 relationships. Zoom 9%.

02 Whole network

02-canvas-whole-network.png

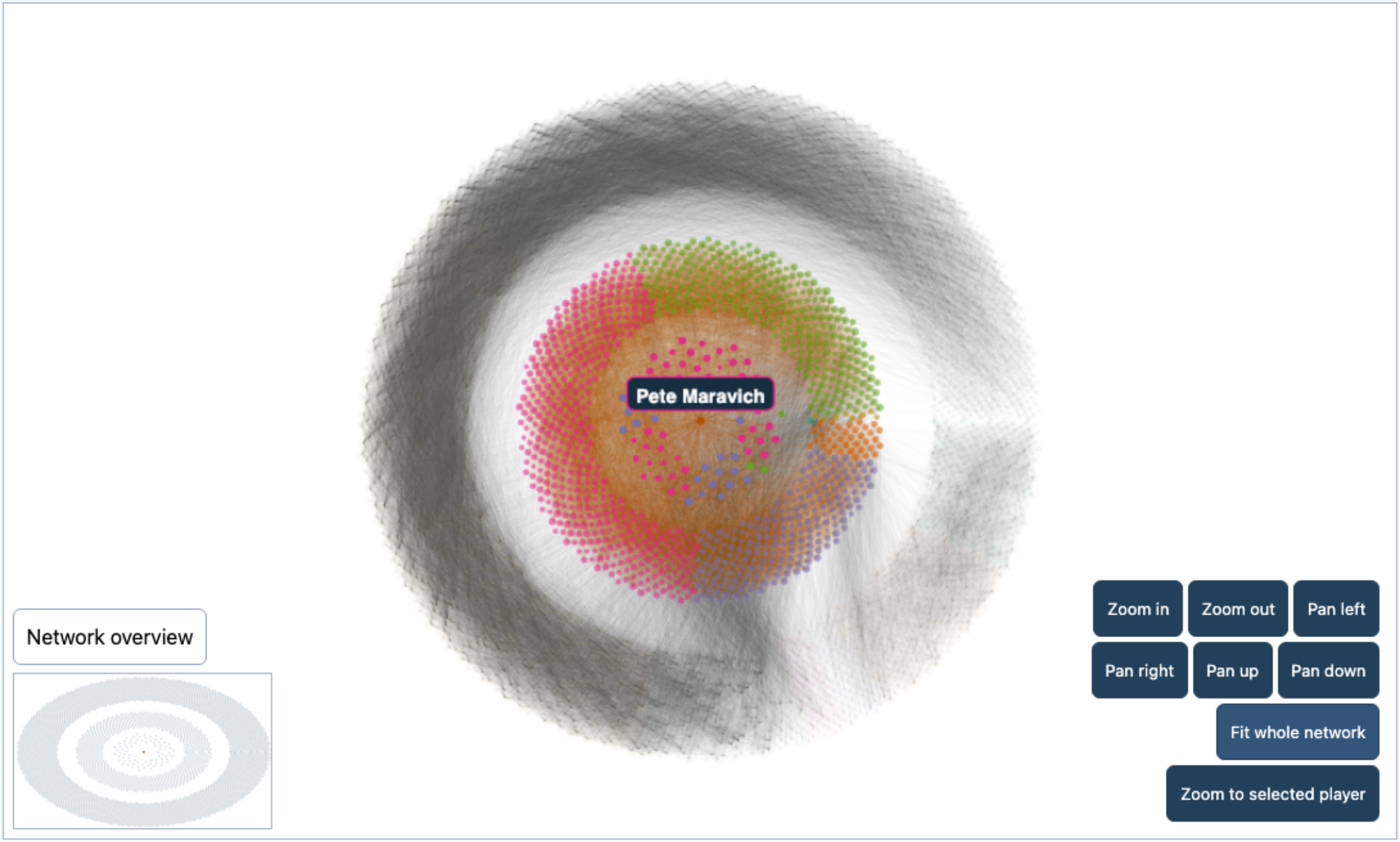
All 5,106 player nodes and 101,395 admitted teammate edges; no active chain.



03 Concentric overview

03-layout-concentric.png

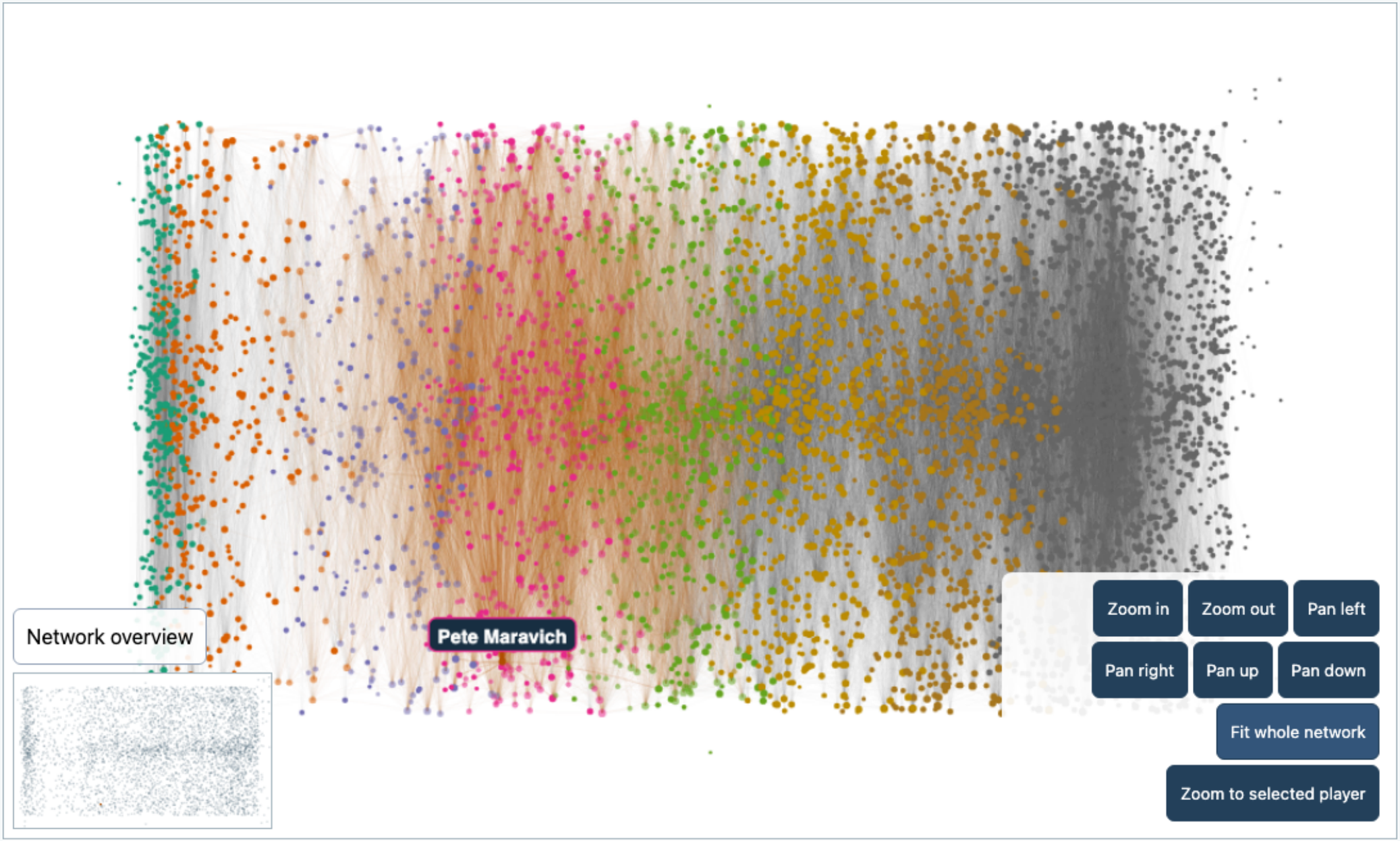
Pete Maravich at the center; direct and two-hop teammates spread across radial bands.



04 Force overview

04-layout-force.png

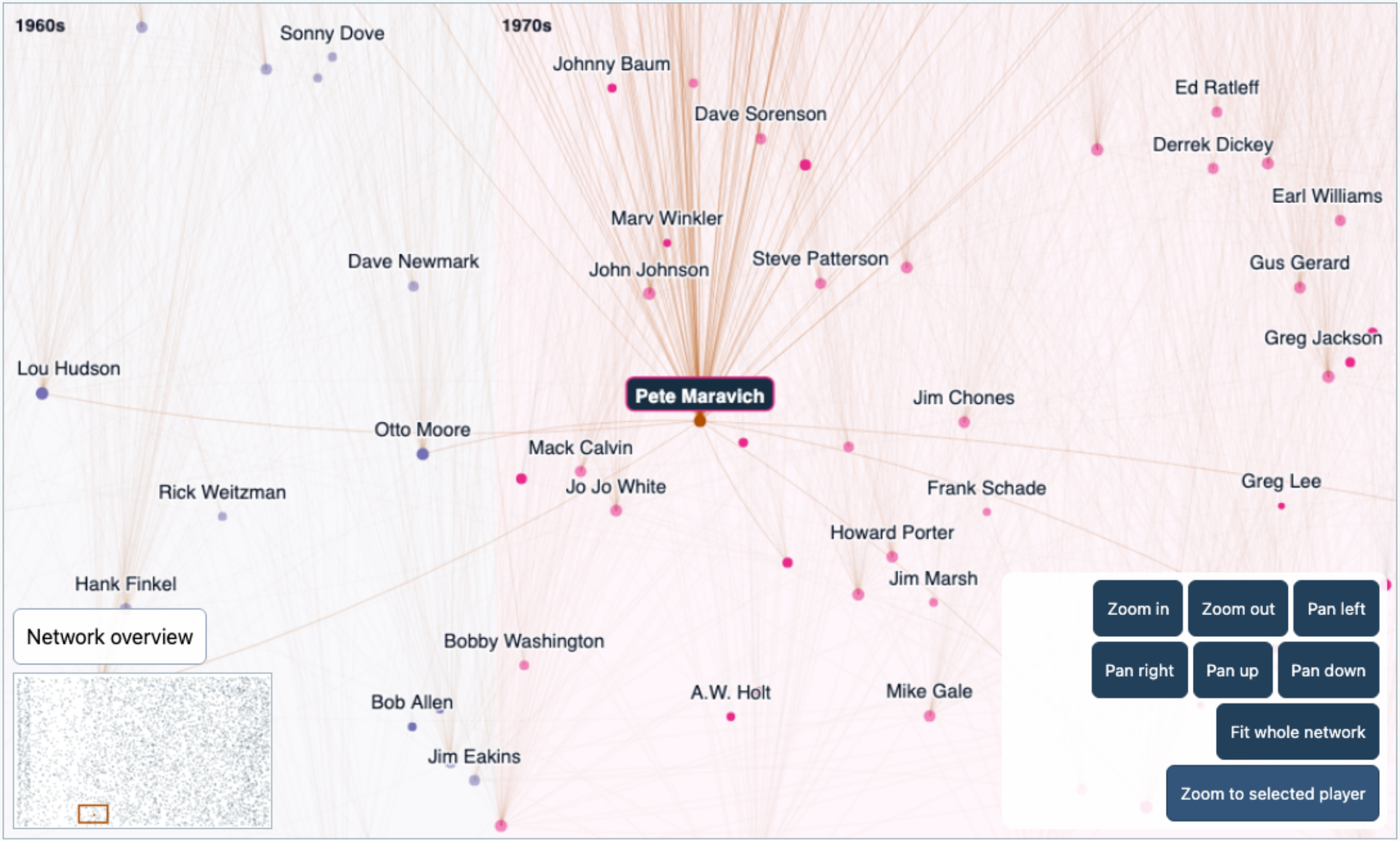
Deterministic force layout seeded from Era lanes, with Pete Maravich selected.



05 Era labels

05-zoomed-labels.png

Readable selected-player badge and ordinary labels with halos and decluttering.



06 Selected player - full page

06-player-selected-focus.png

Pete Maravich selection summary, franchises, filters and focused era canvas.

Graph exploration

Semantic features (Jev): unavailable

All players · all teammate connections

Incomplete dated roster coverage, including 1946–1950 BAA evidence. Links require dated tenure overlap, an identified shared team game, or appearance counts that mathematically prove shared team games. Appearance proofs establish a relationship, but do not invent overlap dates or duration. Season-only membership, ambiguous counts and unresolved transaction brackets never create links. Degrees are exact within this evidenced graph; missing edges may shorten a historical chain or connect currently unreachable players.

Every player and every evidenced connection is loaded. Players are arranged by debut season; colors mark eras. Unknown debut is unclassified context. Select two players for an exact shortest teammate chain.

Player A: Pete Maravich

Player B

Connect players | Random pair | Clear chain

1 match(es). Choose a result explicitly.

Any debut decade | Any franchise | Any debut decade | Any franchise

Era lanes layout | Concentric rings layout | Force-directed layout

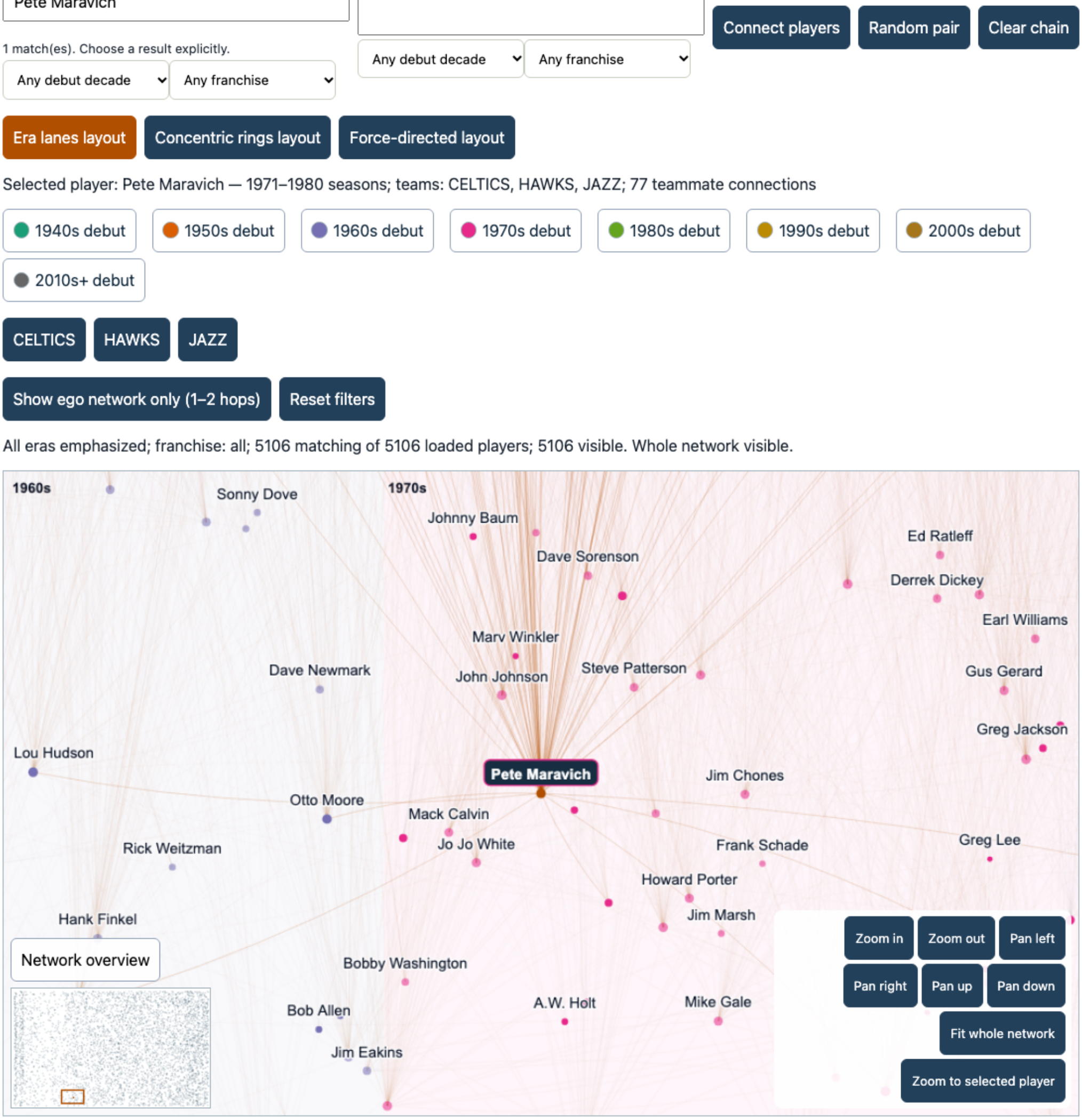
Selected player: Pete Maravich — 1971–1980 seasons; teams: CELTICS, HAWKS, JAZZ; 77 teammate connections

1940s debut | 1950s debut | 1960s debut | 1970s debut | 1980s debut | 1990s debut | 2000s debut | 2010s+ debut

CELTICS | HAWKS | JAZZ

Show ego network only (1–2 hops) | Reset filters

All eras emphasized; franchise: all; 5106 matching of 5106 loaded players; 5106 visible. Whole network visible.



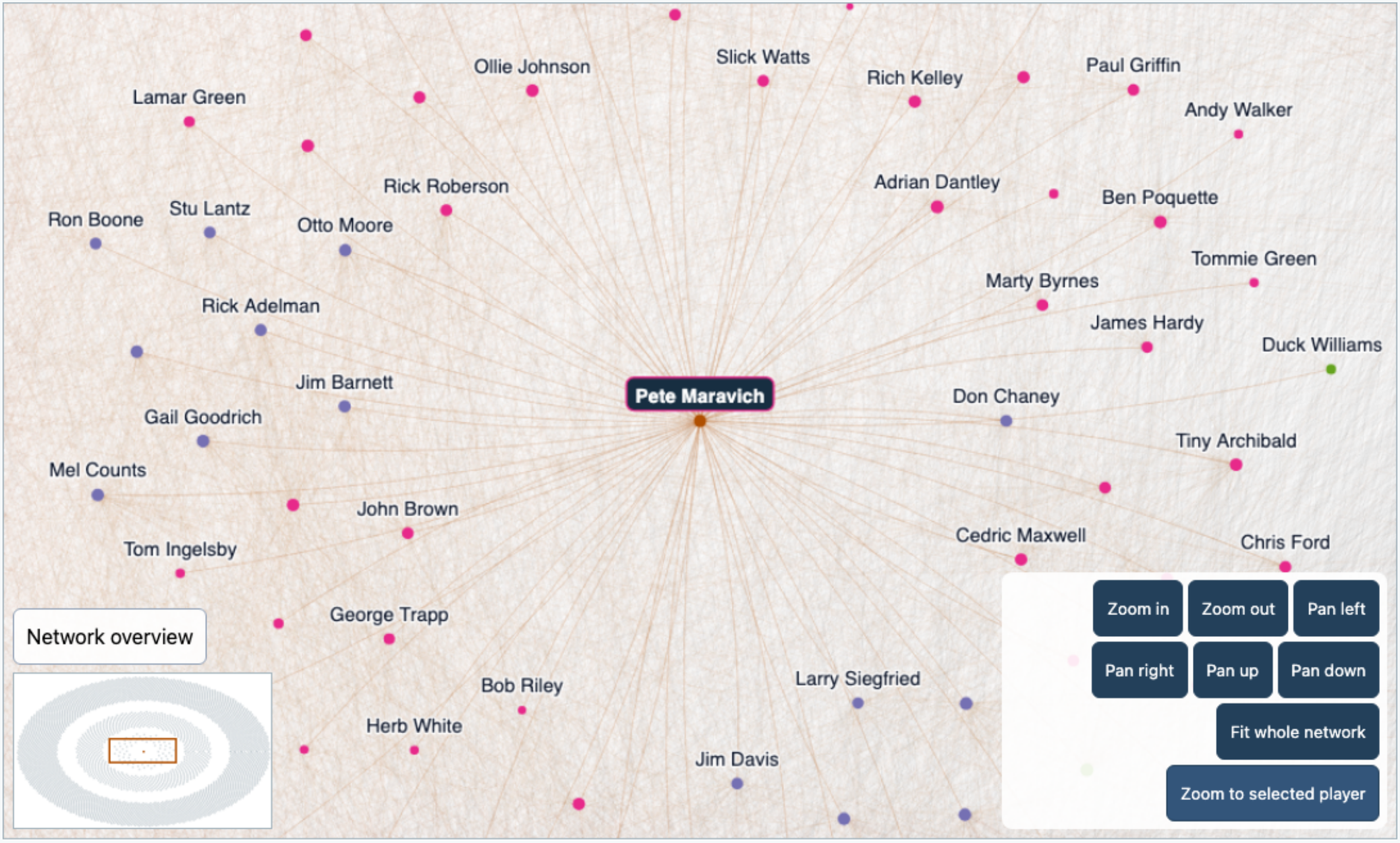
Drag to pan; scroll or pinch to zoom. Arrow keys pan; + and – zoom; f fits the network. Pick a player or click a relationship to inspect evidence.

Canvas ready — 5106 players, 101395 relationships. Zoom 100%.

07 Concentric labels

07-focus-highlight-canvas.png

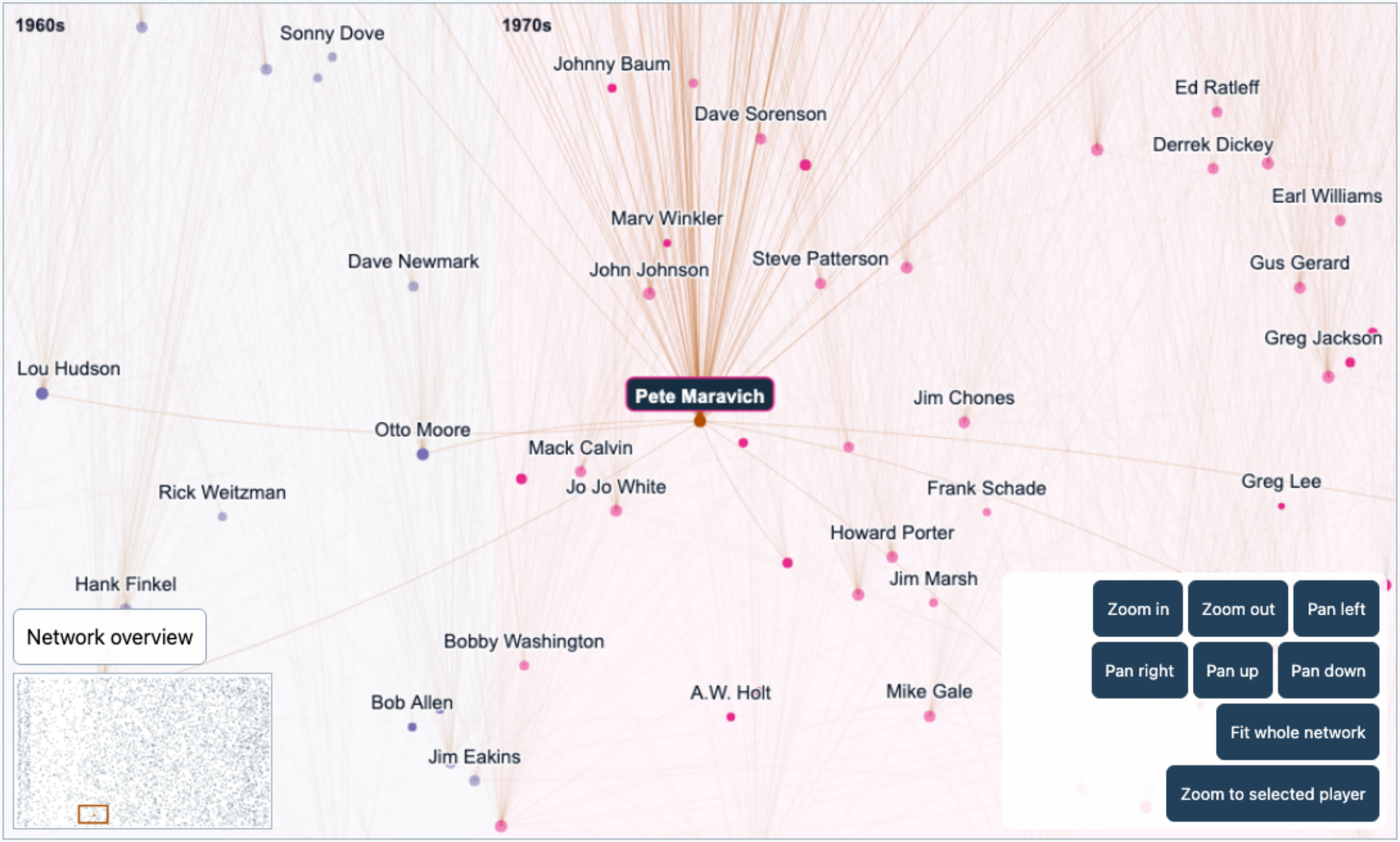
Thin focused curves, selected badge and readable direct teammate context.



08 Docked navigation

08-minimap-and-controls.png

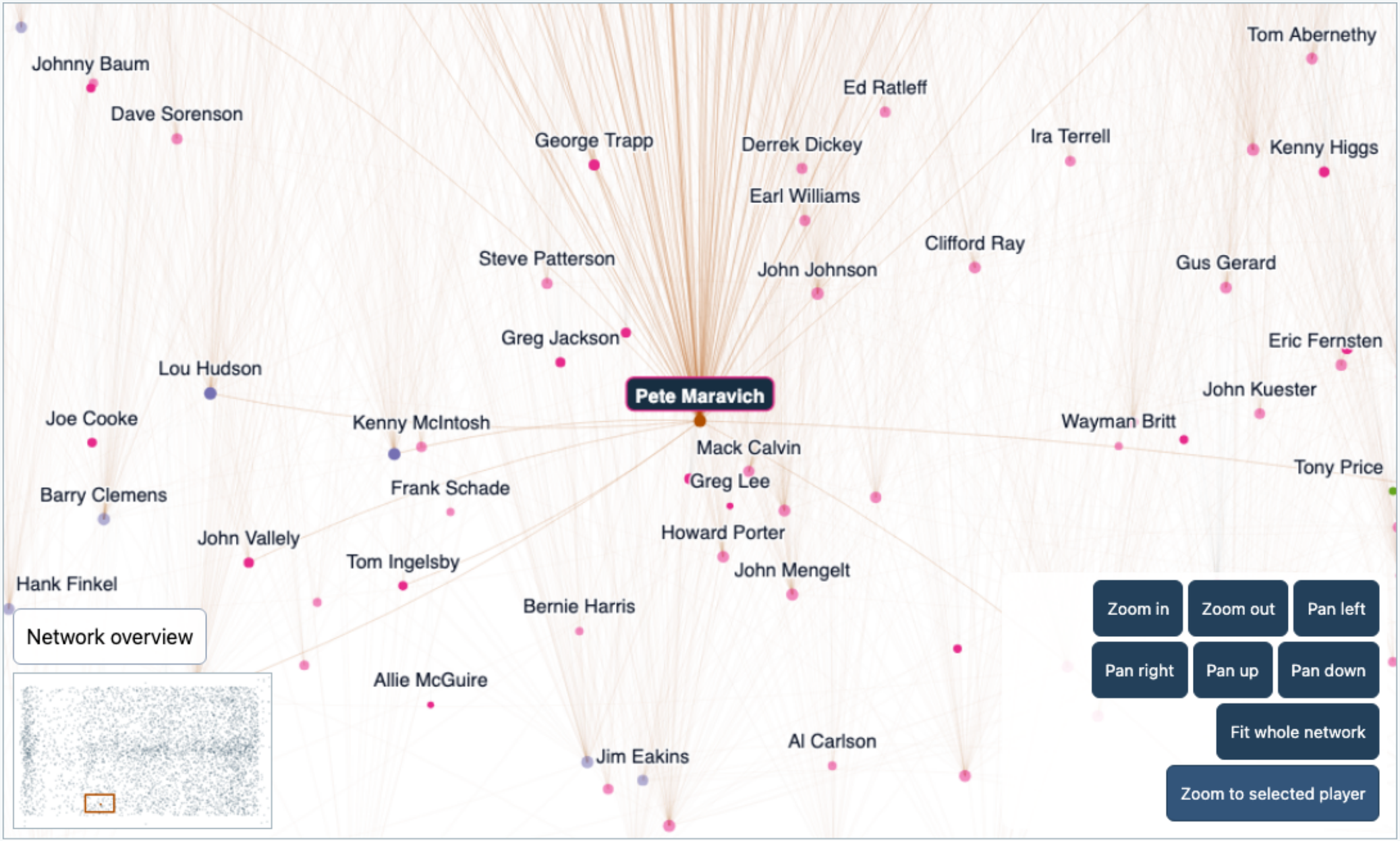
Minimap and its viewport frame at bottom left; camera controls at bottom right.



09 Force labels

09-labels-while-zoomed.png

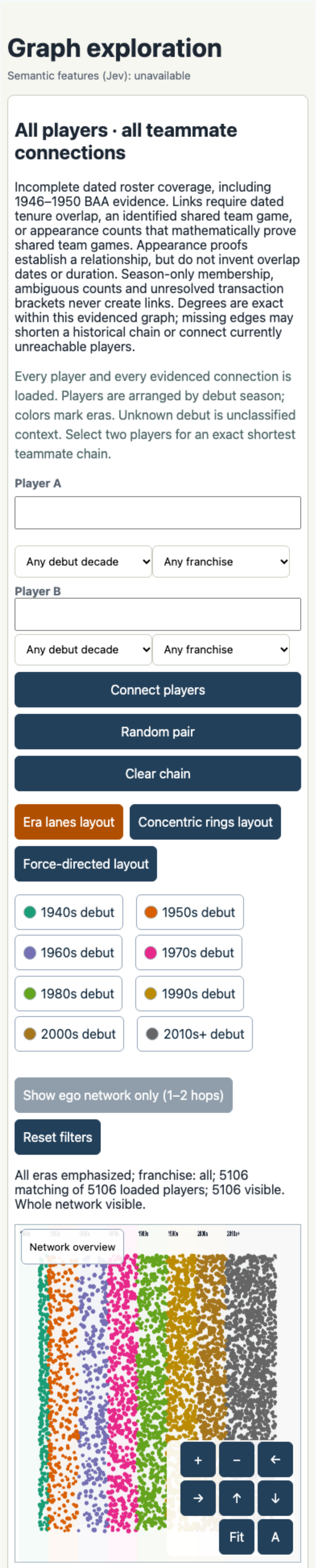
Pete Maravich selected in Force layout at useful label zoom.



10 Phone - whole network

10-mobile-view.png

390-pixel layout with stacked pickers and the minimap collapsed initially.



11 Standalone evidence

11-edge-evidence.png

Direct /edge page for Pete Maravich and Gail Goodrich; count proofs and coverage caveats.

Teammate overlap evidence

Pete Maravich ↔ Gail Goodrich

Teammate relationship admitted by the deterministic graph. The evidence below states which dates or shared games are proved.

Canonical franchise: JAZZ

Pete Maravich ↔ Gail Goodrich

Regular-season appearance proof, season ending 1977: 27 + 73 − 82 = at least 18 shared team game(s).

This proves simultaneous team membership. Exact overlap dates and roster duration are unknown.

Player source records: S2 Player Totals.csv:26189 and S2 Player Totals.csv:26269. Team source record: S2 Team Totals.csv:1422.

Pinned Basketball-Reference-derived statistics, S2 v56

Gail Goodrich: 27/82 appearances · actual season total 82 team games

Pete Maravich: 73/82 appearances · actual season total 82 team games

Counts do not identify which specific games were shared.

Canonical franchise: JAZZ

Pete Maravich ↔ Gail Goodrich

Regular-season appearance proof, season ending 1978: 81 + 50 − 82 = at least 49 shared team game(s).

This proves simultaneous team membership. Exact overlap dates and roster duration are unknown.

Player source records: S2 Player Totals.csv:25828 and S2 Player Totals.csv:25907. Team source record: S2 Team Totals.csv:1400.

Pinned Basketball-Reference-derived statistics, S2 v56

Gail Goodrich: 81/82 appearances · actual season total 82 team games

Pete Maravich: 50/82 appearances · actual season total 82 team games

Counts do not identify which specific games were shared.

Canonical franchise: JAZZ

Pete Maravich ↔ Gail Goodrich

Regular-season appearance proof, season ending 1979: 74 + 49 − 82 = at least 41 shared team game(s).

This proves simultaneous team membership. Exact overlap dates and roster duration are unknown.

Player source records: S2 Player Totals.csv:25470 and S2 Player Totals.csv:25550. Team source record: S2 Team Totals.csv:1376.

Pinned Basketball-Reference-derived statistics, S2 v56

Gail Goodrich: 74/82 appearances · actual season total 82 team games

Pete Maravich: 49/82 appearances · actual season total 82 team games

Counts do not identify which specific games were shared.

Source evidence is incomplete

Missing evidence does not prove a historical non-relationship.

Incomplete dated roster coverage, including 1946–1950 BAA evidence. Links require dated tenure overlap, an identified shared team game, or appearance counts that mathematically prove shared team games. Appearance proofs establish a relationship, but do not invent overlap dates or duration. Season-only membership, ambiguous counts and unresolved transaction brackets never create links. Degrees are exact within this evidenced graph; missing edges may shorten a historical chain or connect currently unreachable players.

▶ Coverage gaps for these players · 25 excluded record(s)

12 Concentric - full page

12-concentric-full-page.png

Toolbar, selected-player context and centered concentric canvas.

Graph exploration

Semantic features (Jev): unavailable

All players · all teammate connections

Incomplete dated roster coverage, including 1946–1950 BAA evidence. Links require dated tenure overlap, an identified shared team game, or appearance counts that mathematically prove shared team games. Appearance proofs establish a relationship, but do not invent overlap dates or duration. Season-only membership, ambiguous counts and unresolved transaction brackets never create links. Degrees are exact within this evidenced graph; missing edges may shorten a historical chain or connect currently unreachable players.

Every player and every evidenced connection is loaded. Players are arranged by debut season; colors mark eras. Unknown debut is unclassified context. Select two players for an exact shortest teammate chain.

Player A

Pete Maravich

1 match(es). Choose a result explicitly.

Any debut decade | Any franchise

Player B

Any debut decade | Any franchise

Connect players | Random pair | Clear chain

Era lanes layout | Concentric rings layout | Force-directed layout

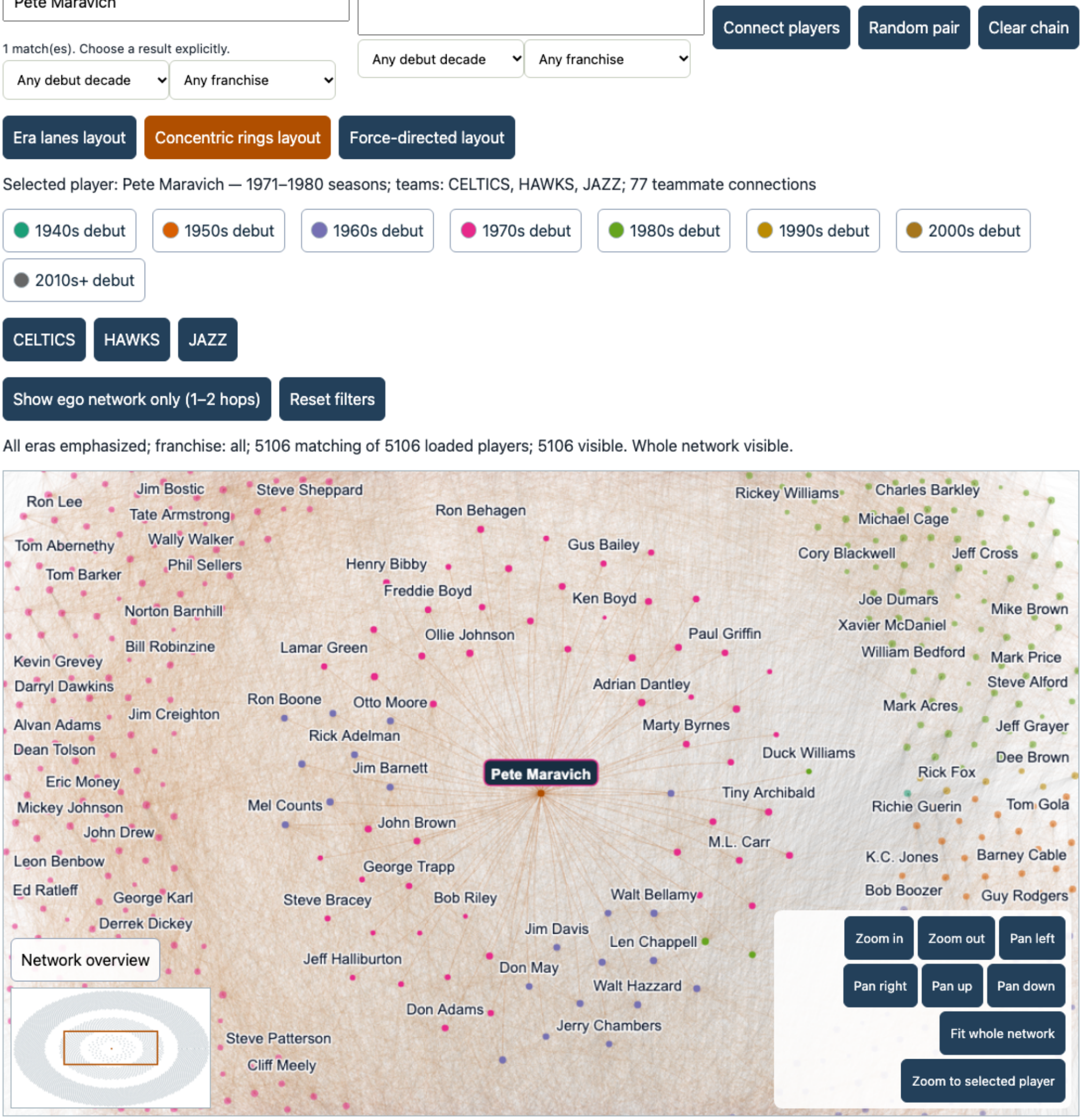
Selected player: Pete Maravich — 1971–1980 seasons; teams: CELTICS, HAWKS, JAZZ; 77 teammate connections

1940s debut | 1950s debut | 1960s debut | 1970s debut | 1980s debut | 1990s debut | 2000s debut | 2010s+ debut

CELTICS | HAWKS | JAZZ

Show ego network only (1–2 hops) | Reset filters

All eras emphasized; franchise: all; 5106 matching of 5106 loaded players; 5106 visible. Whole network visible.



Drag to pan; scroll or pinch to zoom. Arrow keys pan; + and – zoom; f fits the network. Pick a player or click a relationship to inspect evidence.

Canvas ready — 5106 players, 101395 relationships. Zoom 55%.

13 Shortest teammate chain

13-shortest-chain.png

Pete Maravich to Gail Goodrich, degree one; blue chain and ordered evidence breadcrumb.

Graph exploration

Semantic features (Jev): unavailable

All players · all teammate connections

Incomplete dated roster coverage, including 1946–1950 BAA evidence. Links require dated tenure overlap, an identified shared team game, or appearance counts that mathematically prove shared team games. Appearance proofs establish a relationship, but do not invent overlap dates or duration. Season-only membership, ambiguous counts and unresolved transaction brackets never create links. Degrees are exact within this evidenced graph; missing edges may shorten a historical chain or connect currently unreachable players.

Every player and every evidenced connection is loaded. Players are arranged by debut season; colors mark eras. Unknown debut is unclassified context. Select two players for an exact shortest teammate chain.

Player A: Pete Maravich

1 match(es). Choose a result explicitly.

Any debut decade | Any franchise

Player B: Gail Goodrich

1 match(es). Choose a result explicitly.

Any debut decade | Any franchise

Connect players | Random pair | Clear chain

Era lanes layout | Concentric rings layout | Force-directed layout

Selected player: Pete Maravich — 1971–1980 seasons; teams: CELTICS, HAWKS, JAZZ; 77 teammate connections

Shortest teammate chain · degree 1

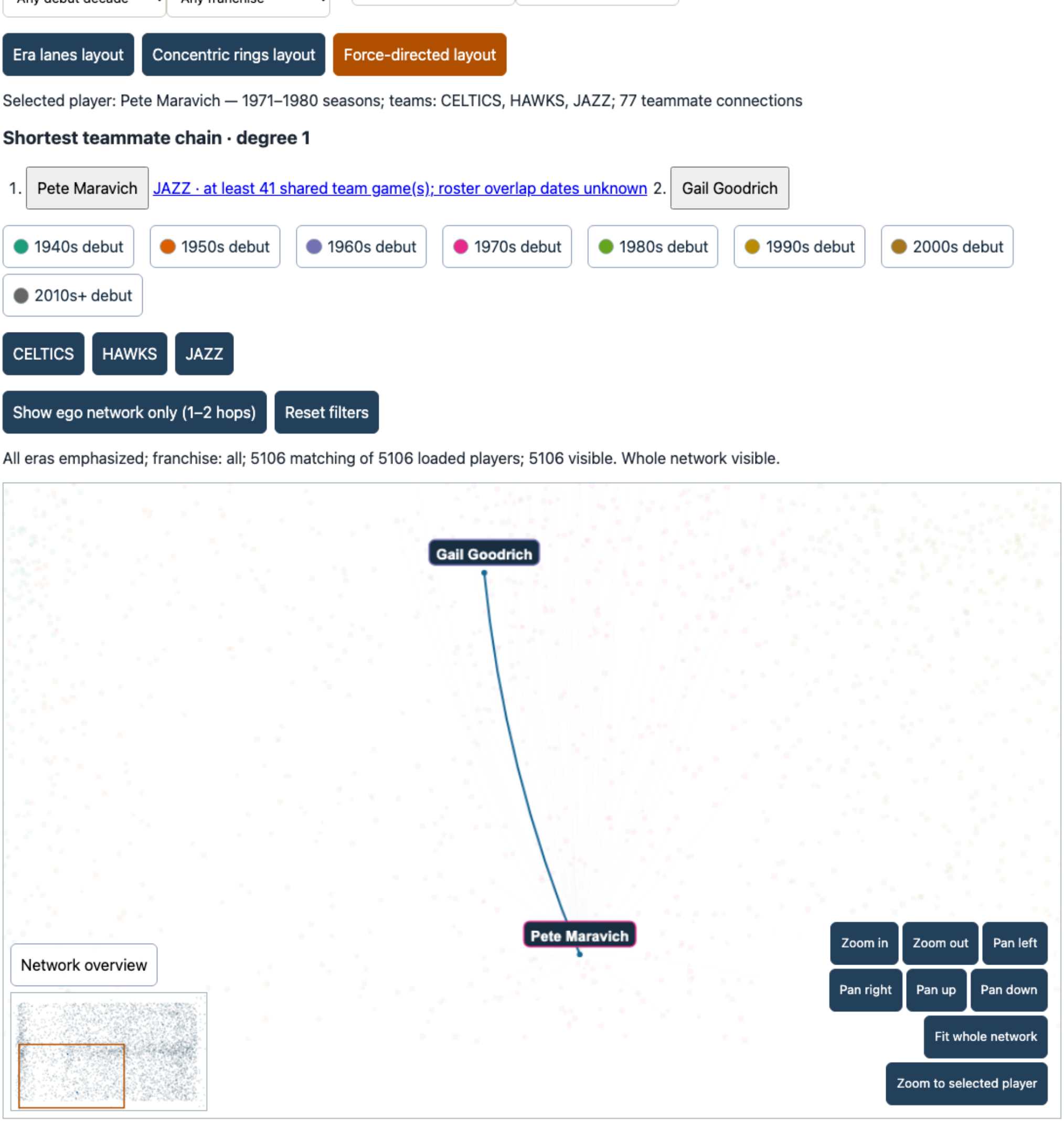
1. Pete Maravich — JAZZ · at least 41 shared team game(s); roster overlap dates unknown — 2. Gail Goodrich

1940s debut | 1950s debut | 1960s debut | 1970s debut | 1980s debut | 1990s debut | 2000s debut | 2010s+ debut

CELTICS | HAWKS | JAZZ

Show ego network only (1–2 hops) | Reset filters

All eras emphasized; franchise: all; 5106 matching of 5106 loaded players; 5106 visible. Whole network visible.



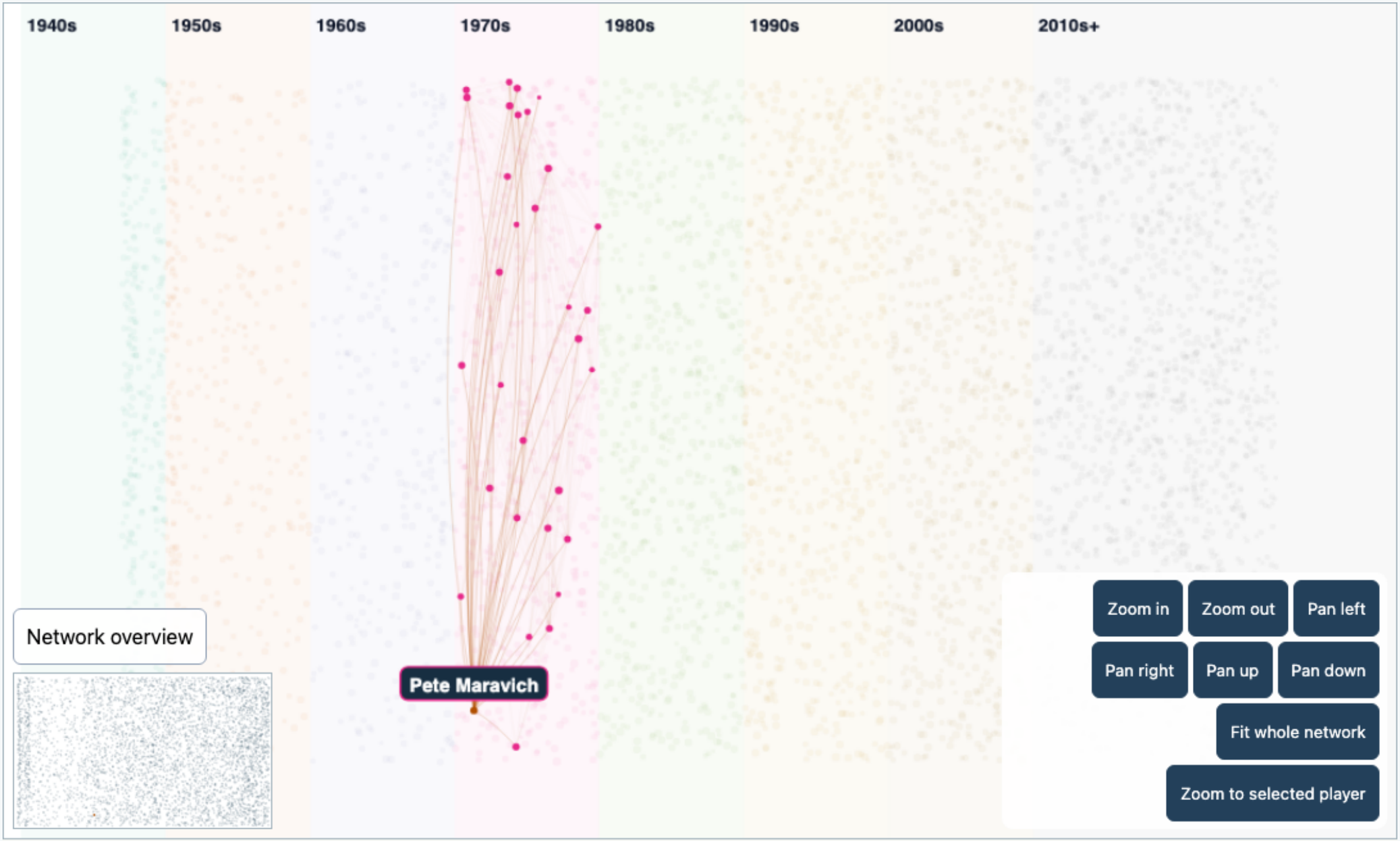
Drag to pan; scroll or pinch to zoom. Arrow keys pan; + and – zoom; f fits the network. Pick a player or click a relationship to inspect evidence.

Canvas ready — 5106 players, 101395 relationships. Zoom 21%.

14 Era and franchise emphasis

14-era-franchise-filter.png

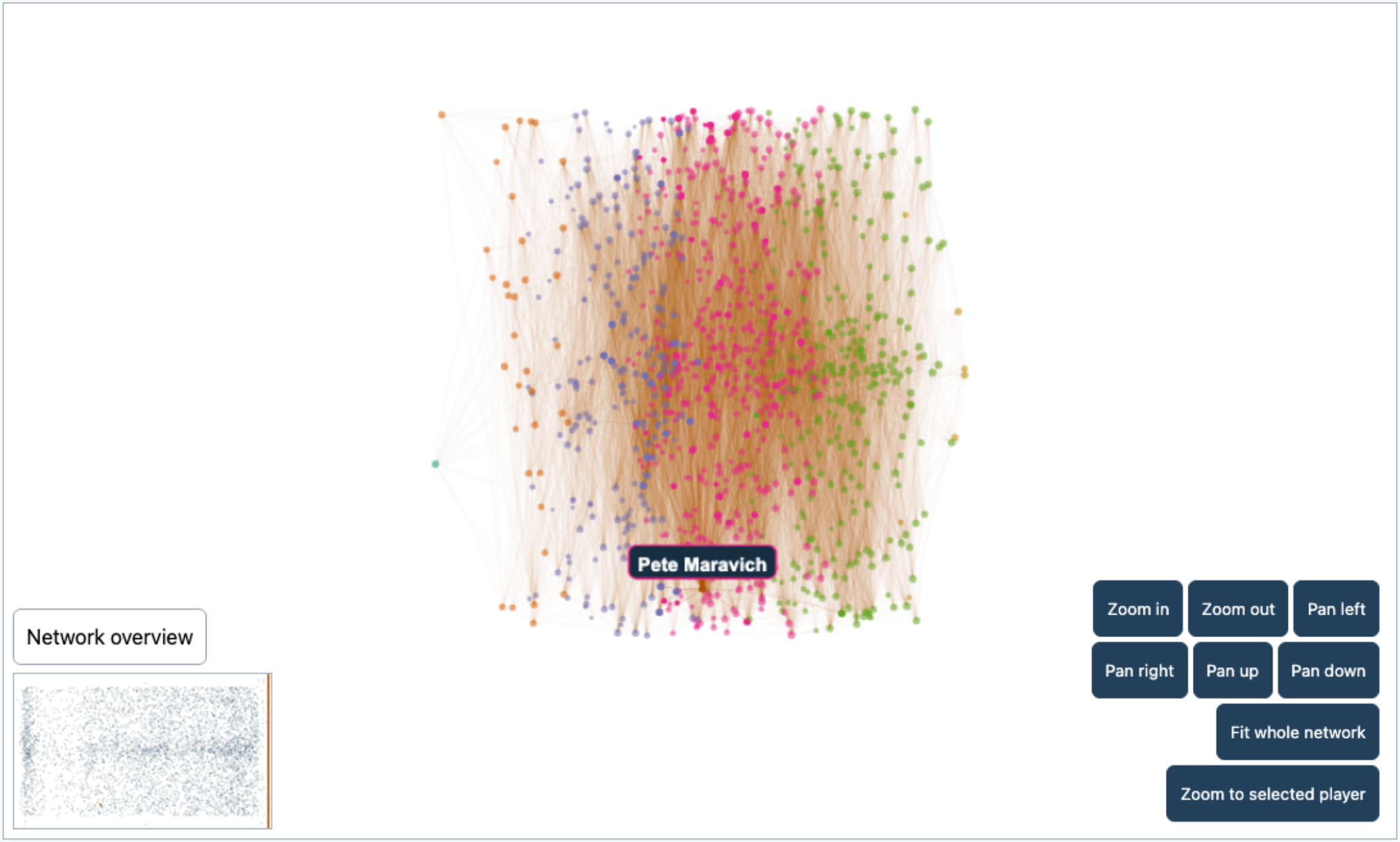
1970s debut emphasis combined with Maravich's admitted JAZZ teammate relationships.



15 Ego network

15-ego-network.png

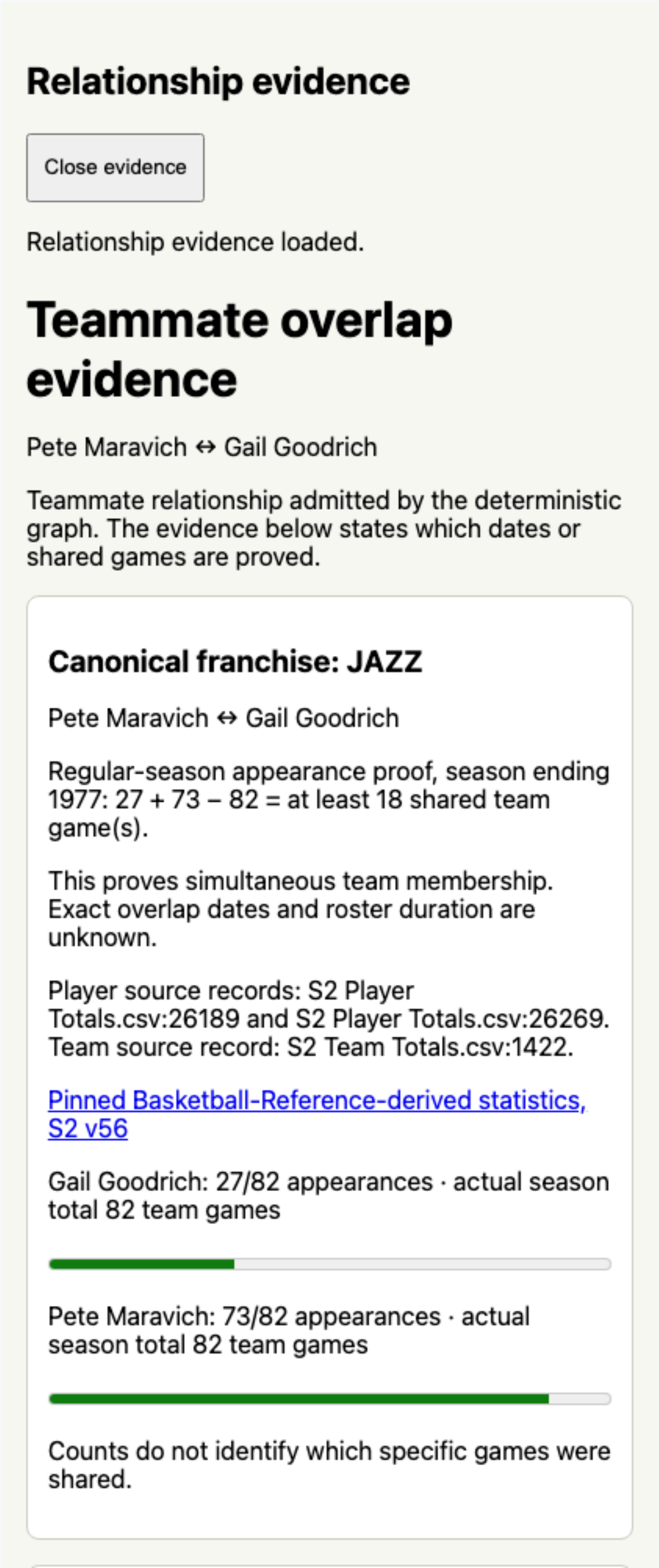
Complete one- and two-hop neighborhood; distant context hidden without unloading graph data.



16 Count-proof drawer

16-count-proof-drawer.png

Actual appearance denominators and arithmetic; exact overlap dates remain unknown.



17 Dated-proof drawer

17-dated-proof-drawer.png

Quincy Acy and Andrew Bogut: MAVERICKS overlap [2016-07-20, 2016-11-18), 121 days.

Relationship evidence

Close evidence

Relationship evidence loaded.

Teammate overlap evidence

Quincy Acy ↔ Andrew Bogut

Teammate relationship admitted by the deterministic graph. The evidence below states which dates or shared games are proved.

Canonical franchise: MAVERICKS

Quincy Acy ↔ Andrew Bogut

Certified dated roster overlap

Overlap: [2016-07-20, 2016-11-18) — 121 day(s); end exclusive.

- Source record t4/tenures.csv:123: Quincy Acy — canonical franchise MAVERICKS; Season 2016–17 (ending 2017); roster tenure [2016-07-20, 2016-11-18) (end exclusive).
  Evidence class
  directly-evidenced
  Membership source
  S2
  Arrival boundary: transaction anchored
  Arrival events: 2016-07-20
  Departure boundary: transaction anchored
  Departure events: 2016-11-18
  Source notes
  None reported
  Source/version manifest
  docs/data/source-manifest.md (S1 v238; S2 v56; BBR request ledger) — Open source/version manifest; Pinned transaction page inventory
- Source record t4/tenures.csv:2602: Andrew Bogut — canonical franchise MAVERICKS; Season 2016–17 (ending 2017); roster tenure [2016-07-07, 2017-02-23) (end

18 Phone - pair controls

18-mobile-toolbar.png

Synthetic A/B chain; stacked controls, readable badges and compact docked navigation.

Graph exploration

Semantic features (Jev): unavailable

All players · all teammate connections

Synthetic fixture; complete within this example.

Every player and every evidenced connection is loaded. Players are arranged by debut season; colors mark eras. Unknown debut is unclassified context. Select two players for an exact shortest teammate chain.

Player A

Player A

Any debut decade | Any franchise

Player B

Player B

Any debut decade | Any franchise

Connect players

Random pair

Clear chain

Era lanes layout | Concentric rings layout | Force-directed layout

Selected player: Player A — Synthetic fixture days 1–11; teams: Green, Red; 2 teammate connections

Shortest teammate chain · degree 1

1. Player A — Red · 2 overlapping roster day(s)
2. Player B

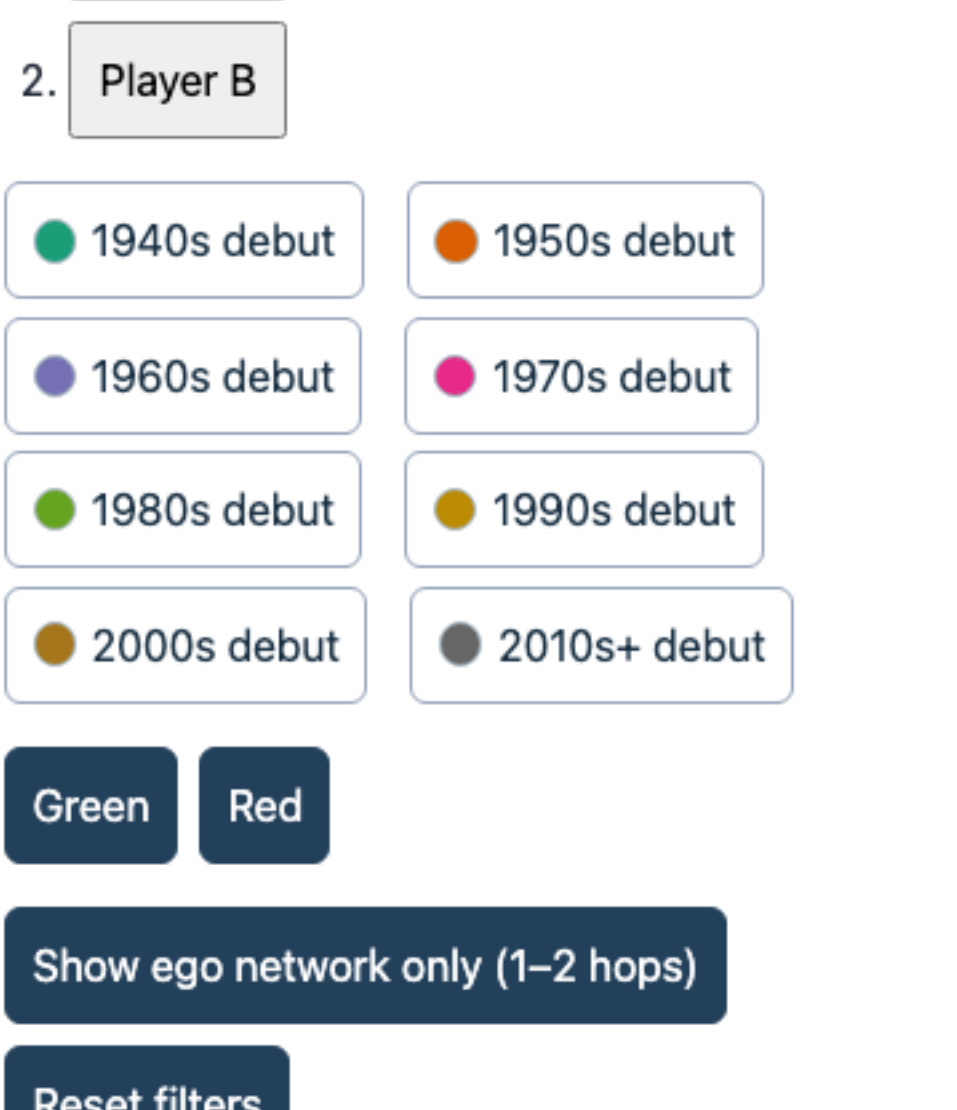
1940s debut | 1950s debut | 1960s debut | 1970s debut | 1980s debut | 1990s debut | 2000s debut | 2010s+ debut

Green | Red

Show ego network only (1–2 hops)

Reset filters

All eras emphasized; franchise: all; 6 matching of 6 loaded players; 6 visible. Whole network visible.



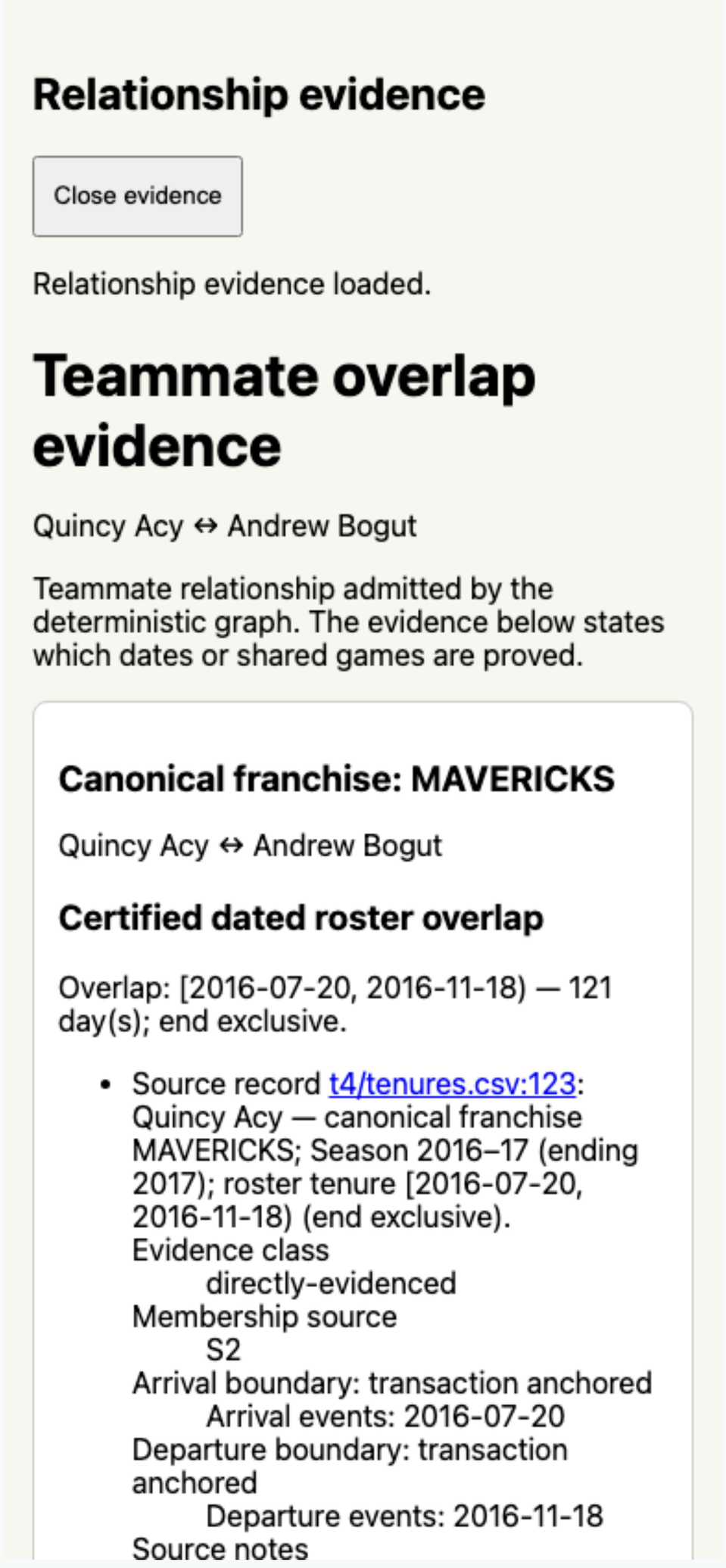
Drag to pan; scroll or pinch to zoom. Arrow keys pan; + and − zoom; f fits the network. Pick a player or click a relationship to inspect evidence.

Canvas ready — 6 players, 5 relationships. Zoom 125%.

19 Phone - evidence drawer

19-mobile-evidence-drawer.png

Real Acy/Bogut dated proof in the viewport-width modal drawer.



20 Reduced-motion final state

20-reduced-motion.png

Settled concentric layout with Pete Maravich selected and reduced motion enabled.



21 Synthetic filtered chain

21-filtered-chain-fixture.png

Known A-B-C chain remains visible while eras are disabled and franchise/ego filters apply.

Graph exploration

Semantic features (Jev): unavailable

All players · all teammate connections

Synthetic fixture; complete within this example.

Every player and every evidenced connection is loaded. Players are arranged by debut season; colors mark eras. Unknown debut is unclassified context. Select two players for an exact shortest teammate chain.

Player A: Player A

1 match(es). Choose a result explicitly.

Any debut decade | Any franchise

Player B: Player C

1 match(es). Choose a result explicitly.

Any debut decade | Any franchise

Connect players | Random pair | Clear chain

Era lanes layout | Concentric rings layout | Force-directed layout

Selected player: Player A — Synthetic fixture days 1–11; teams: Green, Red; 2 teammate connections

Shortest teammate chain · degree 2

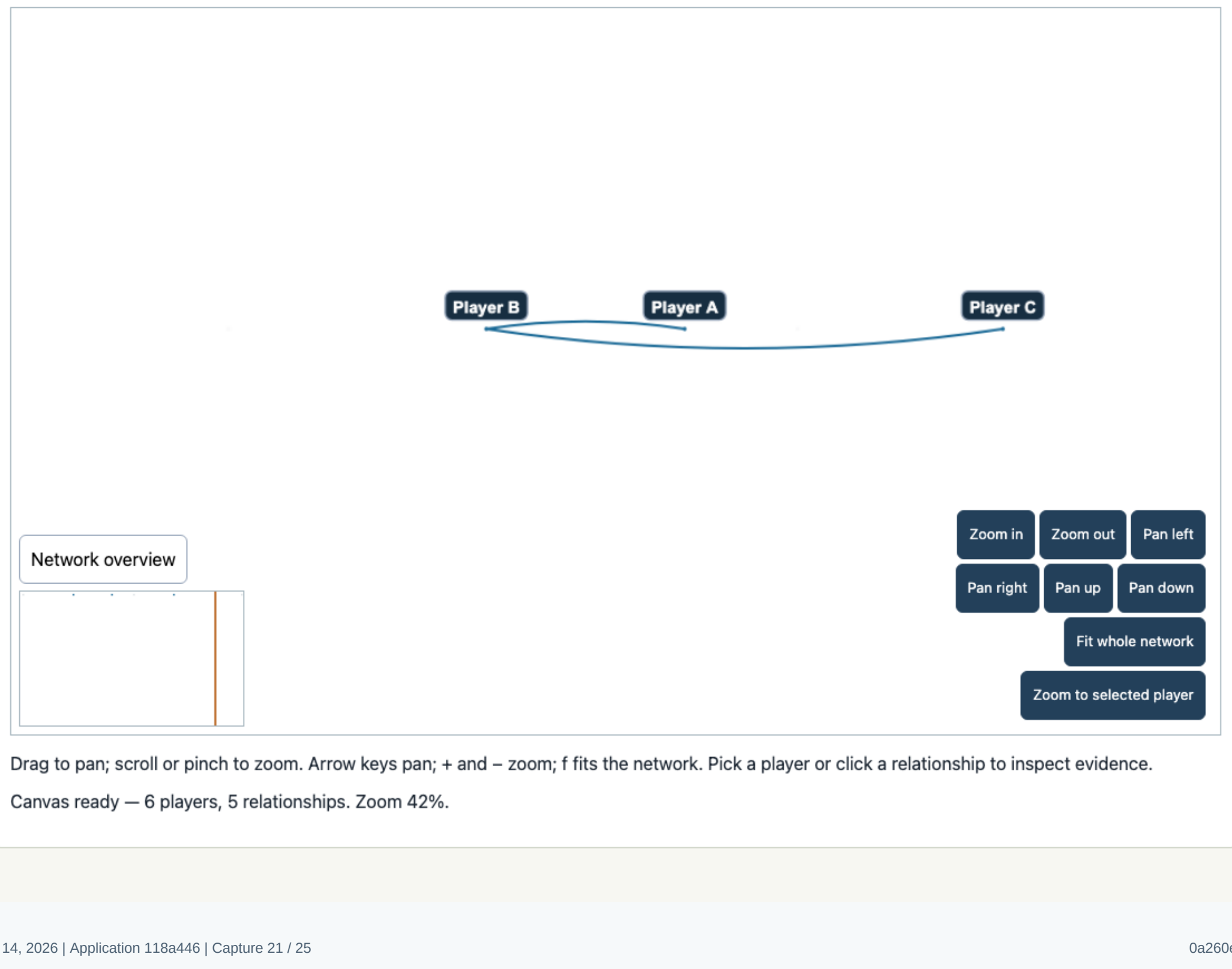
1. Player A — Red · 2 overlapping roster day(s) — 2. Player B — Blue · 1 overlapping roster day(s) — 3. Player C

1940s debut | 1950s debut | 1960s debut | 1970s debut | 1980s debut | 1990s debut | 2000s debut | 2010s+ debut

Green | Red

Show ego network only (1–2 hops) | Reset filters

No eras emphasized; franchise: Red; 0 matching of 6 loaded players; 5 visible. Ego network: complete 1–2 hops; active chain remains included beyond two hops.



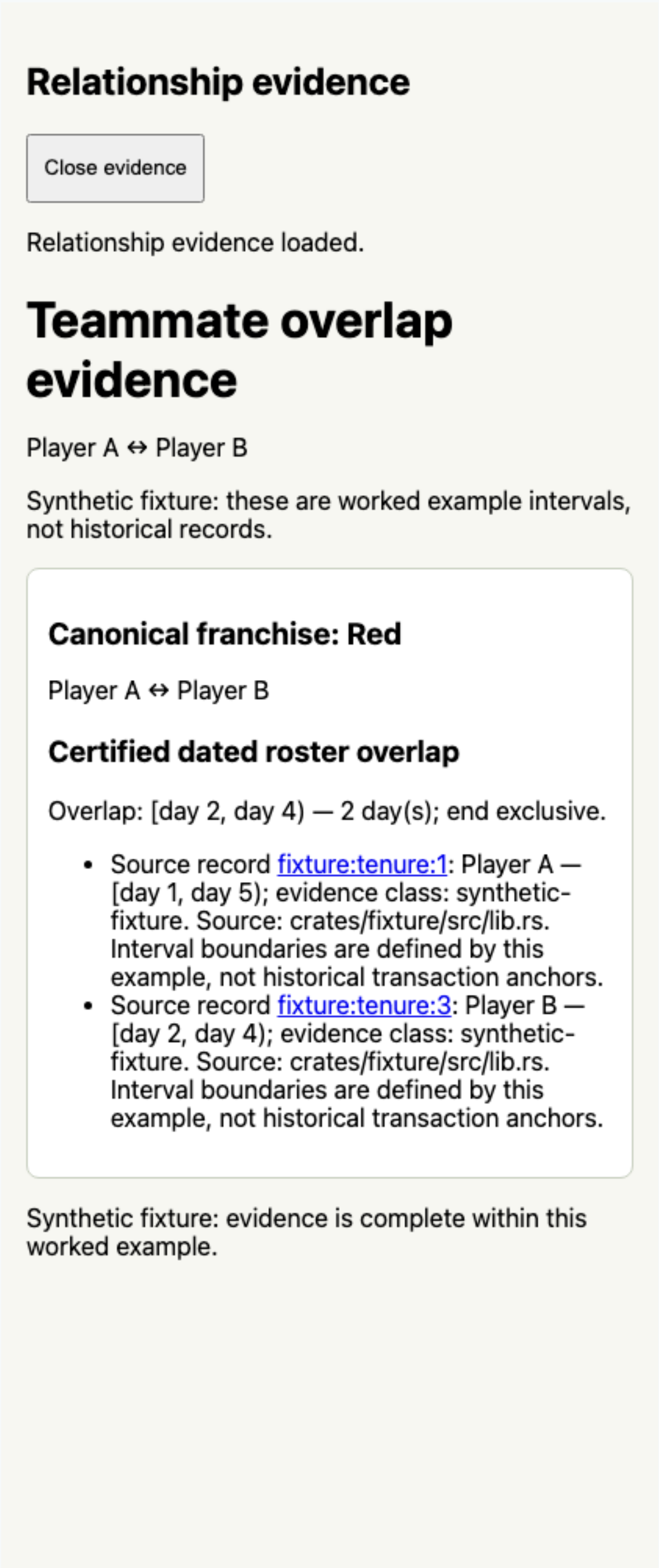
Drag to pan; scroll or pinch to zoom. Arrow keys pan; + and – zoom; f fits the network. Pick a player or click a relationship to inspect evidence.

Canvas ready — 6 players, 5 relationships. Zoom 42%.

22 Synthetic dated proof

22-synthetic-dated-proof.png

A/B worked-example overlap [day 2, day 4); fixture days are not historical dates.



23 Synthetic shortened season

23-shortened-season-proof.png

A 66-game team total guarantees at least 44 shared games from 60 and 50 appearances.

Graph exploration

Semantic features (Jev): unavailable

All players · all teammate connections

Incomplete dated roster coverage, including 1946–1950 BAA evidence. Links require dated tenure o
appearance counts that mathematically prove shared team games. Appearance proofs establish a re
duration. Season-only membership, ambiguous counts and unresolved transaction brackets never c
evidenced graph; missing edges may shorten a historical chain or connect currently unreachable pla

Every player and every evidenced connection is loaded. Players are arranged by debut season; color
context. Select two players for an exact shortest teammate chain.

Player A: Player Alpha — Any debut decade — Any franchise

Player B: Player Beta — Any debut decade — Any franchise

Era lanes layout | Concentric rings layout | Force-directed layout

Selected player: Player Alpha — 1999–1999 seasons; teams: BLUE, RED; 1 teammate connections

Shortest teammate chain · degree 1

1. Player Alpha — BLUE · at least 1 shared team game(s); roster overlap dates unknown — 2. Player

1940s debut | 1950s debut | 1960s debut | 1970s debut | 1980s debut | 2010s+ debut

BLUE | RED

Show ego network only (1–2 hops) | Reset filters

All eras emphasized; franchise: all; 2 matching of 2 loaded players; 2 visible. Whole network visible.

1990s 2000s



Network overview

Zoom in | Zoom out | Pan left | Pan right | Pan up | Pan down | Fit whole network | Zoom to selected player

Drag to pan; scroll or pinch to zoom. Arrow keys pan; + and – zoom; f fits the network. Pick a player or click a relationship to inspect evidence.

Canvas ready — 2 players, 1 relationships. Zoom 125%.

Relationship evidence

Close evidence

Relationship evidence loaded.

Teammate overlap evidence

Player Alpha ↔ Player Beta

Teammate relationship admitted by the deterministic graph. The evidence below states which dates or shared games are proved.

Canonical franchise: RED

Player Alpha ↔ Player Beta

Regular-season appearance proof, season ending 1999: 60 + 50 – 66 = at least 44 shared team game(s).

This proves simultaneous team membership. Exact overlap dates and roster duration are unknown.

Player source records: S2:player:1 and S2:player:2. Team source record: S2:team:1.

Pinned Basketball-Reference-derived statistics, S2 v56

Player Alpha: 60/66 appearances · actual season total 66 team games

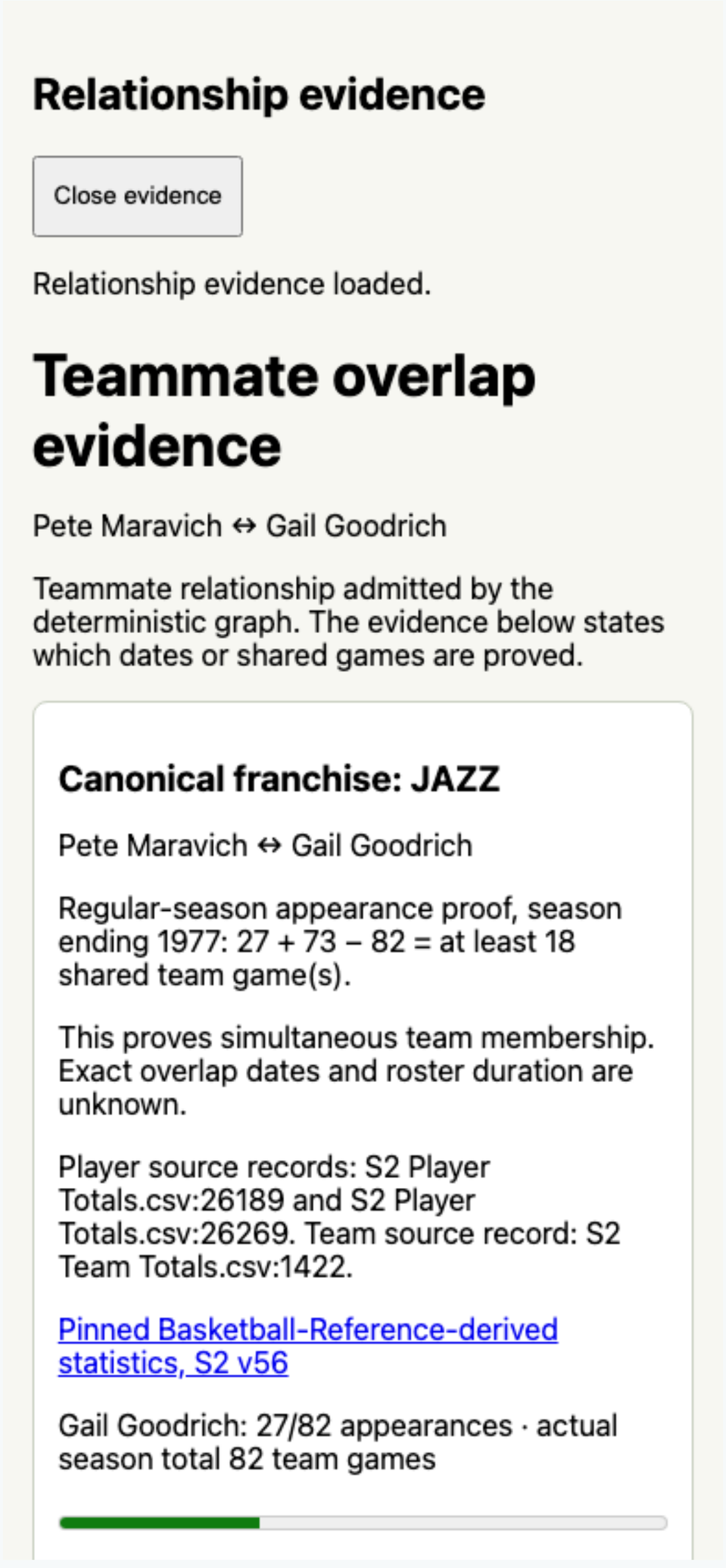
Player Beta: 50/66 appearances · actual season total 66 team games

Counts do not identify which specific games were shared.

24 Phone - count proof

24-mobile-count-proof.png

Maravich/Goodrich proof cards with appearance meters, actual 82-game totals and source records.



25 Official dated game witness

25-game-witness-proof.png

Precious Achiuwa and David Johnson: RAPTORS game 0022100058 on October 27, 2021; original event records.

